

Q-4 (A)
Q-6 OR (A)

Exam Seat No: _____

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B.E. 6th C.E/ I.T (REG/ATKT) EXAMINATION APRIL 2024

Subject Name : Artificial Intelligence

Date: 02/04/2024(Tuesday)

Time: 12.00 pm to 03.00 pm

Subject Code: CT601-N

Total marks: 70

Instructions:

1. Answer each section in separate Answer sheet.
2. All questions are compulsory.
3. Indicate clearly, the options you attempt along with its respective question number.
4. Use the last page of main supplementary for rough work.

Section-I

- Q.1 (A) What is A.I? Where we use and how to solve problems using AI? List out AI characteristics. (5)
- (B) There are two jugs of volume 4 litres and 3 litres. Neither has any measuring mark on it. A pump can be used to fill the jugs with water. How can you get exactly 2 litre of water into the 4-liter jug? Assuming that we have unlimited supply of water. Show state space representation of this problem. (5)
- { (C) What is Hill Climbing? Explain Steepest Ascent Hill Climbing with example. (5)
What is PEAS agent OR in AI? Explain Agent-Environment (5)
(C) Discuss A* algorithm. Compare it with the simple hill climbing search method. (5)
Explain structures of intelligent Agents with its type. (5)
- Q.2 (A) Compare and describe DFS and BFS with example. (5)
- (B) Solve the following Crypt Arithmetic problem. (5)
- DONALD
+ GERALD
ROBERT
- OR
- Q.2 (A) Explain Min-Max procedure with example. (5)
- (B) Solve the following Crypt Arithmetic problem. (5)
- CROSS
+ ROADS
DANGER
- Q.3 (A) What is local maximum, Plateau and Ridge? Describe problem with proper solution. (5)
- ✓(B) Discuss Alpha-beta pruning with alpha-beta cutoffs procedure in Game Playing. (5)
- OR
- Q.3 (A) Which type of problem can be solved by simulated annealing search method? discuss with example. (5)
- (B) Describe AND-OR graphs. (5)
- Differentiate betⁿ forward & backward and clarify Example*

Section-II

- Q.4 (A) Discuss Cut, Fail and Repeat ^{predicates} predicts in prolog. (5)
(B) Define Ontology. Discuss RDF with example. (5)
(C) Write a prolog program to find factorial of given number. (5)

OR

- (C) Write a prolog program to find Fibonacci series for given number. (5)
- Q.5 (A) Explain Bayesian Networks with example. (5)
(B) Consider following axioms: (5)
1. Every child loves every candy.
2. Anyone who loves some candy is not a nutrition fanatic.
3. Anyone who eats any pumpkin is a nutrition fanatic.
4. Anyone who buys any pumpkin either carves it or eats it.
5. John buys a pumpkin.
6. Lifesavers is a candy.
Prove using resolution – “If John is a child, and then John carves some pumpkin.”

OR

- Q.5 (A) Explain different approaches to knowledge representation. (5)
(B) Consider following axioms: (5)
1. Anyone whom Mary loves is a football player.
2. Any student who does not pass does not play.
3. John is a student.
4. Any student who does not study does not pass.
5. Anyone who does not play is not a football player.
Prove using resolution – “If John doesn’t study, Mary doesn’t love John.”

- Q.6 (A) Describe semantic web and frames with example. (5)
(B) What is fuzzy logic? How it can be used in Decision Making under uncertainty for given problem. (5)

OR

- Q.6 (A) Discuss the algorithm of ^{predicate} ~~predict~~ logic resolution. (5)
(B) Define and discuss belief network in AI. (5)


~~Explain PCA representation for an Agent with the~~
~~example.~~